

Abstract

In the face of climate change, it becomes increasingly important to examine the ways in which consistent exposure to higher temperatures can impact organisms. Chemoreception is one of the main mechanisms through which crabs sense the world, and there have been many studies conducted to investigate the impact of pH upon chemoreception. Far less work has been done on any possible correlation between temperature and chemoreception. Using *Hemigrapsus oregonensis* as our experimental subject, we held a control group at 14C and a treatment group at 24C for three weeks. We tested their ability to detect both food and predators once a week and measured their metabolic activity with a resazurin assay. After analyzing our data in R, we found little evidence to suggest that temperature plays a role in chemoreception in *H. oregonensis*.

Introduction

Crabs utilize chemoreception for a variety of cues, including food, predators, and mates (Ritschoff 2015). Much work has been done investigating the impact of pH upon the chemoreception of many species of crabs (Khodikian et al. 2025, Richardson et al. 2021), but there has been comparatively little on the effects of temperature. With an increasing number of heatwave events in both marine and terrestrial systems, such as the 2014 blob and the 2021 heat dome, it is vital to explore the ways extreme heat can impact species, particularly intertidal species that can be affected by both (Chen et al. 2025, Miner et al. 2025).

For this study, we wanted to explore two main questions. How does thermal stress impact chemoreception in shore crabs? Do elevated temperatures affect reception and response time for crabs experiencing food and predator-related chemical cues? To answer these questions, we tested the following hypotheses:

1. *H. oregonensis* held in control conditions will find food faster than elevated temperature treatments.
2. Control *H. oregonensis* crabs will respond to predator chemical cues with more antennule flicking (higher rates of water sampling) relative to crabs held at elevated temperatures.
3. *H. oregonensis* held at elevated temperatures will show elevated respirometry results, indicating a heightened metabolic rate.

Methods

For this experiment, we used 12 total *H. oregonensis* split into two groups. Six were randomly selected for the control group and held at 14C for the duration of the study and the other six were placed in the treatment group and held at 24C. Individuals were marked with a number for data recording purposes.

Food cues

To test responsiveness to food cues, individuals were placed in a simple maze designed to break line of sight to the food offered (we used small pieces of fish). Subjects were fasted prior to testing. They were placed in the maze and given a minute to adjust before we added the food and timed how long it took them to locate the food and begin eating. Due to time constraints, we gave them each a maximum of five minutes to locate the food and considered them as having failed to find the food if they exceeded this maximum.

Predator cues

To study response to predator cues, subjects were placed in a tank with a barrier that allowed water to pass between the partitioned sections but broke line of sight between the sections. Subjects were given a minute to adjust to the tank, and then we placed [Green crab] in the other section of the tank. We recorded the reactions of the individuals for one minute to quantify antennule activity in the presence of a potential predator. We broke this down into quantitative and qualitative measurements: amount of time in that minute spent flicking, and level of intensity of the flicking.

Respiration

To investigate the impact on metabolic activity of our subjects, we used resazurin assays over a period of one hour. Resazurin is a redox-sensitive dye that is reduced by cellular respiration, which then acts as a proxy for metabolic activity. By measuring the fluorescence of the resulting assays, we can approximate the metabolic activity of our subjects. Subjects were placed in trays with enough resazurin to cover them and assays were drawn at three separate times: T_0 prior to placing crabs in the resazurin, T_1 after crabs had been in the resazurin for 30 min, and T_2 30 minutes after T_1 (a total of one hour in the resazurin). We then measured the fluorescence of the assays.

Data collection and analysis

All data was compiled using Excel and analyzed using R with the package ggplot2.

Results

Food cues

Crabs held in elevated temperatures responded more quickly to a food cue, with a lower average time taken to locate food. The difference between the two treatment groups was not significant, with both groups having similar average times (*Figure 1*). Across both weeks of testing, only three crabs took the full five minutes allotted to them, two in the control group and one in the treatment group.

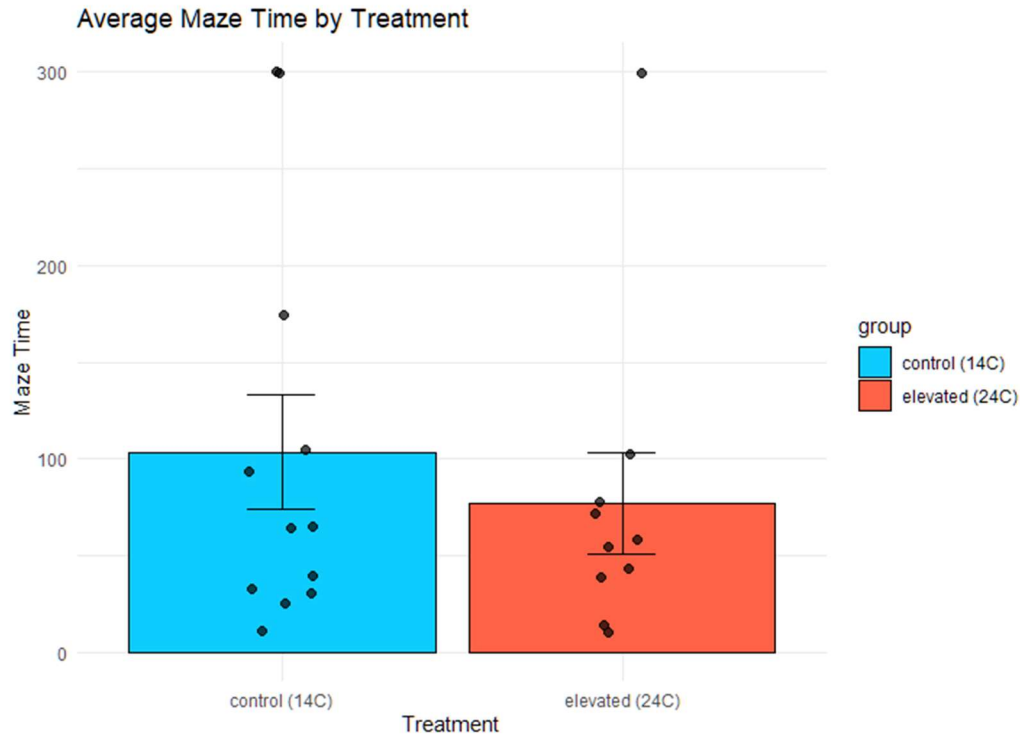


Figure 1: This graph shows the average time (seconds) it took for crabs to find food in the maze with individual data points overlaid on the bars. On average, the control group took longer to find food, but it is not a significant difference.

Predator cues

Both groups had generally similar responses to the presence of predators. The amount of time spent flicking their antennules (Figure 2) shows both groups had far more variance in week one than in week two, but both groups spent the majority of the minute sampling the water. In week two, the variance narrowed until almost all subjects were spending upwards of 80% of the time sampling the water. The treatment group spent slightly less time than the treatment group, but not by a significant margin.

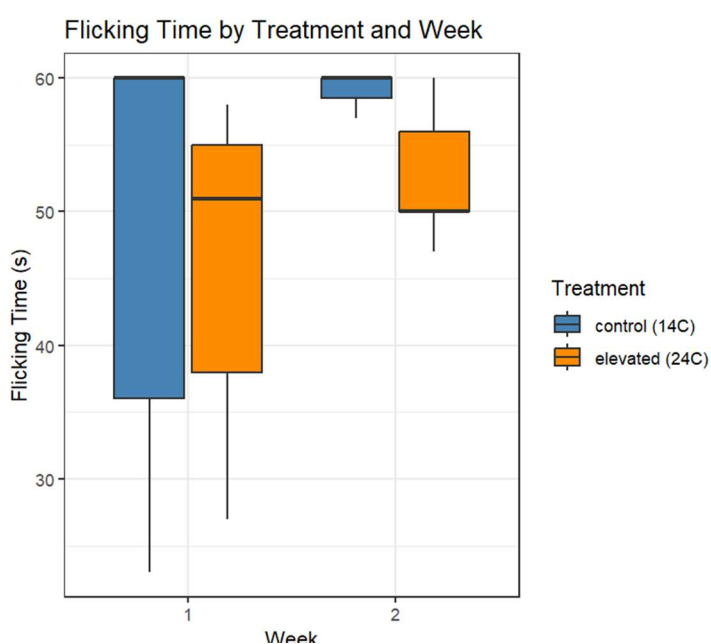


Figure 3: This graph shows the amount of time subjects spent flicking their antennules while in the presence of a potential predator.

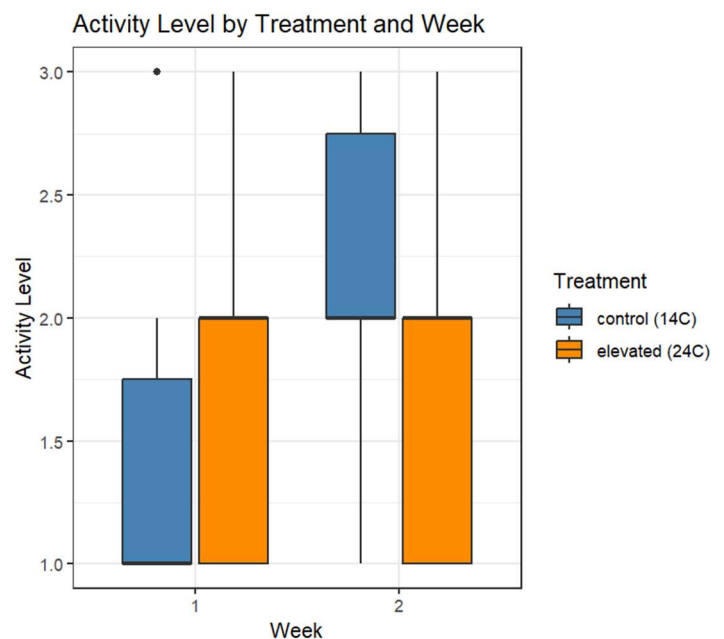


Figure 2: This graph displays a more qualitative measurement describing the intensity of observed antennule flicking, with 1 being low intensity and 3 being high intensity.

Similar to the amount of time spent sampling the water, the intensity of that sampling was similar across both groups (*Figure 3*). The treatment group had the same activity level both weeks, but the treatment group saw an overall increase in their activity in week two. The differences between the groups were not substantial.

Respiration

The crabs held in elevated conditions initially respired more compared to the ones held in control temperatures. However, respiration rate subsequently decreased for crabs in the both treatment groups after week 1.

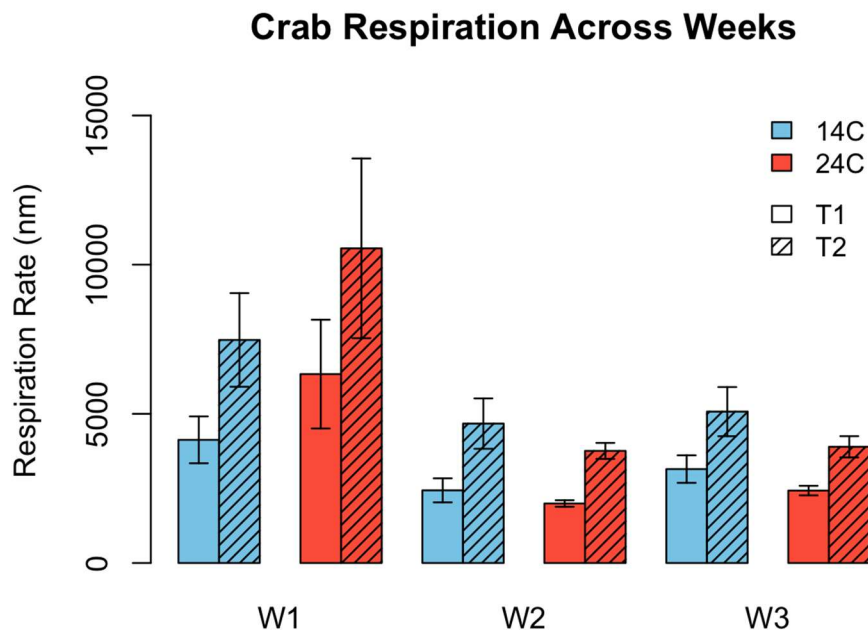


Figure 4: This graph shows the respiration rate of subjects. T1 is the first reading taken from our crabs, and T2 is the second reading in the same week.

Discussion

There was not a significant difference between our control and treatment groups across any of our recorded metrics.

Maze times were similar, with crabs in the treatment group outperforming the control group on average. Before we had finalized our respiration results, we thought this could be due to the fact that higher temperatures can increase metabolic rates in many animals (Clarke & Fraser 2004). Once we had our respirometry results, though, we found that both groups had similar rates. A more likely explanation is natural variation that would be less apparent with a larger sample size.

We also failed to find a significant difference in subject reaction to predator cues. Both groups had evident reactions to the presence of a predator with high activity levels and near-constant sampling of the water with antennules.

After three weeks in the elevated temperature tank, the experimental crabs showed signs of acclimatization, as demonstrated by respiration rates comparable to the control group. The elevated respiration in week one for the treatment group could also be attributed to crab 12, which had the highest respiration rate and died between weeks 1 and 2. Considering that the control group's respiration also decreased between weeks 1 and 2, though, acclimatization is likely at play for all groups. The high respiration in week 1 could very well be attributed to stress induced by the initiation of the experiment itself before acclimatizing to laboratory conditions.

It could also very well be that the thermal stress our treatment group was exposed to was simply within its natural tolerance range, and in order to see any effects on chemoreception or metabolic rate, we would need to hold them at a higher temperature. Other studies have shown that intertidal species, in particular, have wide thermal tolerances that may allow for them to adapt to persistent higher temperatures (Madeira et al. 2018).

For a future study, it would be interesting to examine the differences in the metabolic rate of wild crabs against that of fasted crabs in a laboratory setting. Brockington and Clarke found that wild sea urchin metabolic rates in the summer were less influenced by temperature than they were by feeding and reproduction, and that fasted individuals in a lab did not have significant changes to their metabolic rate even in warmer water (2001). Considering that our subjects were fasted for the duration of the study (apart from the food they received in the maze), this is an avenue worth investigating.

References

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